

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: Alcohol Intermediary Herbs-Infused Oil Composed of Oregano (Origanum vulgare), Lemongrass (Cymbopogon citratus), Neem (Azadirachta indica), and Butterfly pea (Clitoria Ternatea)	
Proponent Name:	Princess Lydia S. Fuentes
Academic Rank:	Associate Professor V
Co-Authors (with academic rank):	
Campus/Satellite College/Address:	Dayao Satellite College
Telephone/Cell Phone/Fax Number:	+639306532341
E-mail Address:	pls Fuentes@capsu.edu.ph

Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
---	--

(4) Site of implementation	
Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☒ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
---	---

(7) Sector/Commodity Product Development	(8) Discipline/Field of Specialization Chemistry
--	--

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 2 of 10

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

In the recent years, there has been an increased public demand for safe, effective, and cost-efficient herbal oil. The interest is in the use of aromatic plants, which have a wide range of applications due to their diverse properties such as antioxidant, expectorant, diuretic, antimicrobial and antifungal. With the increased public demand for safe, effective, and cost-efficient herbal oil. This study aimed to prepare and characterize alcohol intermediary herbs-infused oil using locally grown plants oregano, lemongrass, neem, and butterfly pea. While it is known in the art the process of extracting oil through infusion, alcohol intermediary herb-infused oils have a longer shelf-life due to the use of alcohol, which lowers the possibility of microbial growth in the oil. This study showed that the extraction and the process of producing oil from oregano, lemongrass, neem and butterfly pea is acceptable within the standards based on the results of some physical test. All treatments complied to the given standards as to refractive index, density, pH, and brix index. Refractive index ranges from 1.450 to 1.453, density ranges 0.70-0.82 g/mL, brix index ranges 64.45 – 65.60%. However, although the results on the lower pH can increase the solubility of oils, only T11 (pH = 5.5) is within the skin's pH range of 5.5. Making T11 (5% oregano, 5% lemongrass, 5% neem, and 5% butterfly pea) the most acceptable formulation so far in the coverage of this study.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)
essential oil, herbal oil, alcohol-intermediary, refractive index, brix value, density, pH

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Oil from plant extracts has been used as a remedy for joint, muscle, and skin inflammation in Southeast Asia since ancient times and even in the present time. Herbal oil from various herbal ingredients has always been used as alternative medicine in many parts of Asia. Currently, many scientific studies are trying to prove the therapeutic effects of herbal oils.

Oregano oil is an essential oil extracted from the oregano plant that is claimed to have healing properties. Research shows oregano oil has antimicrobial, antiviral, and antifungal properties. In addition, compounds in oregano oil are also a potent antioxidant, anti-inflammatory, antidiabetic, and cancer-suppressor agents (Sharififard, et al., 2005). Lemongrass is a tropical, grassy plant used in cooking and herbal medicine. Lemongrass oil can be extracted from the leaves and stalks of the lemongrass plant, and it has been used by healthcare providers to treat digestive problems and high blood pressure. In fact, lemongrass essential oil is a popular tool in aromatherapy to help relieve stress, anxiety, and depression. Among many other benefits, lemongrass oil has antibacterial properties, has antifungal properties, has anti-inflammatory properties, has antioxidant properties, may help prevent gastric ulcers or relieve nausea, may help ease diarrhea, may help reduce cholesterol, may help regulate blood sugar and lipids, may act as a pain reliever, may help relieve stress and anxiety, and may help relieve headaches and migraine (Shah, et al, 2011). Neem oil can be derived from the fruits and seeds of the neem tree. This type of oil contains several compounds, including fatty acids and antioxidants, that can benefit the skin. Neem oil is rich in fatty acids, such as palmitic, linoleic, and oleic acids, which help support healthy skin. Further, its leaves contain plant compounds called flavonoids and polyphenols, which have antioxidant, anti-inflammatory, and antibacterial properties (Alzohairy, 2016). Butterfly pea extract is very useful in the skin and hair care industry due to the high concentration of naturally occurring polyphenols (proanthocyanidin or condensed tannins) and flavonoids (Jeyaraj, et al, 2021). Proanthocyanidin, in particular, helps stimulate the natural production of collagen and elastin. Flavonoids, on the other hand, are antioxidants that help reducing inflammation,

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 3 of 10

redness, stress, and skin aging. Recent studies have confirmed that lactic acid fermented Butterfly pea flower has a particularly strong free-radical scavenging and anti-inflammatory ability. It inhibits redness, itching, allergies, and irritation to the skin.

In the Philippines, many patented technologies have combined different oils and extracts from herbal plants to produce herbal oils for topical use (Santos, 2019; Bolivar, 2017; Doloriel & Doloriel, 2018). However, the composition and the process of preparing the oils in the context of this study is considered novel over the existing prior arts/ studies.

Generally, this study aimed to prepare and characterize alcohol intermediary herbs-infused oil composed of Oregano (*Origanum vulgare*), Lemongrass (*Cymbopogon citratus*), Neem (*Azadirachta indica*), and Butterfly pea (*Clitoria Ternatea*).

Specifically, this study aimed to:

1. prepare alcohol intermediary herbs-infused oil composed of Oregano (*Origanum vulgare*), Lemongrass (*Cymbopogon citratus*), Neem (*Azadirachta indica*), and Butterfly pea (*Clitoria Ternatea*); and
2. determine the characteristics of alcohol intermediary herbs-infused oil composed of Oregano (*Origanum vulgare*), Lemongrass (*Cymbopogon citratus*), Neem (*Azadirachta indica*), and Butterfly pea (*Clitoria Ternatea*) in terms of color, scent/fragrance, texture, pH, density, refractive index, and brix value.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The experimental method composed of:

1. Development of the product
 - 1.1 Preparation of raw materials
Oregano leaves, lemongrass, neem leaves, butterfly pea flowers, alcohol, carrier oil
 - 1.2 Preparation of other materials
canning jar and lid, parchment paper, blender, stainless steel sieve, cheesecloth, filter paper, blender
 - 1.3 Formulation of composition

Table 1. Different formulation of composition of herbal oil.

Treatment	Alcohol	Oregano	Lemongrass	Neem	Butterfly pea	Carrier oil
A	10%	20%				70%
B	10%		20%			70%
C	10%			20%		70%
D	10%				20%	70%
E	10%	10%	10%			70%
F	10%	10%		10%		70%
G	10%	10%			10%	70%

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 4 of 10

H	10%	10%	10%	70%		
I	10%	10%	10%	70%		
J	10%		10%	10%	70%	
K	10%	5%	5%	5%	5%	70%

1.4 Alcohol intermediary herbs-infused oil making process

- Drying of herbs
- Grinding of dried herbs
- Herbs and alcohol mixture preparation
- Infusing the mixture for 24 hours
- Blending the mixture
- Adding carrier oil
- Blending the mixtures
- Heating the mixtures in a medium heat
- Straining
- Transferring the mixture to sterilized glass container

2. Characterization of the product

Parameters measured: color, scent/fragrance, pH, density, refractive index. Brix %

Color and scent/fragrance were based on observed qualitative characteristics of the oil by 50 evaluators. On the other hand, pH which refers to the acidity and alkalinity of oils was measured using a pH meter. Density which refers to the mass per unit volume was measured by measuring the mass and volume of the sample. Refractive index is a unique number that designates how the oil responds and bends light. Brix value refers to the concentration of soluble solids and is equivalent to the percentage by weight. Both refractive index and brix value were measured using a refractometer.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 10

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Characterization of the Herbal Oil

The herbal oil was evaluated for color, scent, and texture as shown in Table 2. Scent is described using the seventeen (17) descriptors to describe an herbal or aromatherapy scent (Sharmeen, et al., 2021).

Table 2. Qualitative Characteristics of the Herbal Oil

Treatment	Replication 1			Replication 2			Replication 3		
	Color	Scent	Texture	Color	Scent	Texture	Color	Scent	Texture
1	Yellow	Minty	Heavy	Light Yellow	Minty	Light	Light Yellow	Minty	Light
2	Light Yellow	Minty	Light	Light Yellow	Minty	Light	Light Yellow	Minty	Light
3	Light Yellow	Sharp/pungent	Heavy	Light yellow	Sharp	Light	Light yellow	Sharp	Light
4	Yellow to green	Herbaceous	heavy	Yellow to green	Herabaceous	Light	Yellow to green	Herabaceous	Light
5	Yellow to green	Citrus	Slightly heavy	Light Yellow	Citrus	Slightly heavy	Light Yellow	Citrus	Slightly heavy
6	Yellow to green	Herbaceous	Light	Yellow to Green	Herbaceous	Slightly heavy	Yellow to Green	Herbaceous	Slightly heavy
7	Yellow to green	Minty	Heavy	Light Yellow	Minty	Heavy	Light Yellow	Minty	Heavy
8	Yellow to green	Sweet	Light	Yellow to Green	Sweet	Slightly Heavy	Yellow to Green	Sweet	Slightly Heavy
9	Yellow	Sweet	Heavy	Yellow	Sweet	Slightly heavy	Yellow	Sweet	Slightly heavy
10	Yellow	Sharp/pungent	Slightly heavy	Yellow	Sharp	Slightly heavy	Yellow	Sharp	Slightly heavy
11	Yellow	Sharp/pungent	Heavy	Yellow	Sharp	Slightly heavy	Yellow	Sharp	Slightly heavy

Results showed that the color of the oil is in yellow to green spectrum for all treatments. As to scent, it can be observed that the aromatic scent depends on the plant used. All treatments exhibit desirable scent such as minty, herbaceous, sweet, citrus, and sharp. The minty scent is associated with Oregano while the Herbaceous/Citrus scent can be associated to lemon grass. The Sharp/pungent scent can be associated to neem leaves, while the sweet scent is due to butterfly pea. In terms of texture it appears that the first four treatments (T1, T2, T3, and T4) are light. Treatments T5, T6, T8, T9, T10, T11 are slightly heavy. It was observed that these treatments are more viscous compared to the first four treatments. However, Treatment, T7 appears to be heavy or thick in texture.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 10

pH Level

The different treatments in three replications undergone pH testing using a handheld portable pH meter. pH level refers to the acidity and alkalinity of oils. pH affects the solubility of essential oils when using a solubilizer. A lower pH can increase the solubility and make dissolving oils easier.

Table 3. pH Level of the different oil composition

Treatment	pH Level			
	Replication 1	Replication 2	Replication 3	Mean
1	5.7	5.5	5.6	5.6
2	3.9	5.3	4.5	4.6
3	4.3	5	4.65	4.65
4	4.1	4.1	4.1	4.1
5	4.5	4.2	4.2	4.3
6	4.4	4.3	4.3	4.33
7	4.6	4	4.3	4.3
8	4.5	3.9	4.2	4.2
9	6.7	4.2	5.4	5.43
10	5.6	4	4.8	4.8
11	5	6	5.5	5.5

Table 3 showed the results of the pH testing of the different composition of oils. It can be concluded that all treatments are acidic, with pH level ranging from 4.1-5.6 indicating that the different composition is unsuitable for microbial activity and growth. Among all treatments, T4 (pH=4.1) is the most acidic. Treatment 11 (pH=5.5) is the most acceptable for skin (pH=5.5) since any product has a pH that is way out of our skin's pH range may cause irritation.

Refractive Index

The different treatment was tested for refractive index using a refractometer. A sample is transferred into the glass slide of the refractometer. In order to carry out determination of the refractive index of the oils, the measuring prism was opened and a drop of the oil is placed on it and closed with the small cover plate. Refractive index describes how the oil responds and bends light. However, refractive index is only a qualitative test of purity of essential oils and does not give percentage purity.

Table 4. Refractive Index of the different oil composition.

Treatment	Refractive Index			
	Replication 1	Replication 2	Replication 3	Mean
1	1.453	1.453	1.453	1.453
2	1.451	1.453	1.452	1.452
3	1.452	1.452	1.452	1.452
4	1.452	1.453	1.453	1.453
5	1.451	1.453	1.452	1.452
6	1.451	1.453	1.452	1.452
7	1.452	1.452	1.452	1.452
8	1.451	1.453	1.452	1.452
9	1.452	1.452	1.452	1.452
10	1.447	1.452	1.45	1.450
11	1.453	1.453	1.453	1.453

Based on the results on the refractive index of the different oil composition, it can be inferred that all treatments' refractive index is within the standards. The refractive index range (1.450 to 1.453) is

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 7 of 10

comparable to rose oil (1.450-1.484), FAO, Rome (2001) and to cottonseed oil (1.450-1.480), International Gem Society (2024). According to Armando (2009), the higher the refractive index value of essential oil, the oil will be purer or better and the lower the refractive index value, the oil will get worse.

Density

Density is a measure of mass per unit volume. It is a fundamental physical property and can be used together with other properties to characterize oils. Measuring the density of oils may be helpful to ensure that it matches the product specified and remains in acceptable condition for further use. It is also an essential step in determining fluid compatibility.

Table 5. Density of the different oil composition.

Treatment	Density (g/mL)			Mean
	Replication 1	Replication 2	Replication 3	
1	0.66	0.98	0.82	0.82
2	0.68	0.78	0.73	0.73
3	0.72	0.68	0.7	0.70
4	0.7	0.78	0.74	0.74
5	0.7	0.78	0.74	0.74
6	0.68	0.72	0.7	0.70
7	0.72	0.76	0.74	0.74
8	0.7	0.72	0.71	0.71
9	0.68	0.86	0.77	0.77
10	0.74	0.76	0.75	0.75
11	0.7	0.88	0.79	0.79

The results on the density (0.70-0.82 g/mL) of the different oil composition are less than 1 g/mL. The density of most essential oils extracted from plants are less than 1g/mL. Essential oils are often associated with the components within. High density will affect the content in it, and vice versa if essential oils have a low specific gravity, the content in them will be less (Hikmah, 2017).

Brix

Brix value refers to the concentration of soluble solids (salts, sugar, proteins, fatty acids, etc.) and is equivalent to the percentage by weight. In this study, brix value is measured using a refractometer. Brix is an important in controlling microorganisms. The higher the Brix value is, the less water is available for microbial growth, as the sugars are bound to the free water.

Table 6. Brix Value of the different oil composition.

Treatment	Brix %			Mean
	Replication 1	Replication 2	Replication 3	
1	65.1	65.0	65.05	65.05
2	64.3	64.8	64.55	64.55
3	64.4	64.5	64.45	64.45
4	65.5	65.0	65.25	65.25
5	65.1	65.0	65.05	65.05

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 8 of 10

6	65.4	65.8	65.6	65.60
7	65.2	65.9	65.55	65.55
8	65.4	65.2	65.3	65.30
9	64.7	65.2	64.95	64.95
10	65.2	65.7	65.45	65.45
11	65.1	65.8	65.45	65.45

The values of the Brix index determined for all treatments ranged between the lowest 64.45% (T3) and the highest 65.60% (T6). Therefore, a 100 g of oil sample has 64.45 – 65.60 g of sugar and other dissolved solids and 64.45-65.60 g of water. Results complied within the standards though a little lower brix value with the existing known oils such as peppermint (92.15%) and mint oil (92.10%), Mieso & Befu (2020).

Table 7. Physical Properties of the Different Oil Composition

<i>Treatment</i>	<i>pH</i>	<i>Refractive Index</i>	<i>Temperature (°C)</i>	<i>Density (g/mL)</i>	<i>Brix (%)</i>
1	5.6	1.4530	29.05	0.82	65.05
2	4.6	1.4520	29.55	0.73	64.55
3	4.65	1.4524	30.25	0.70	64.45
4	4.1	1.4526	30.65	0.74	65.25
5	4.35	1.4517	31.00	0.74	65.05
6	4.35	1.4522	31.05	0.70	65.60
7	4.3	1.4519	31.15	0.74	65.55
8	4.2	1.4521	31.20	0.71	65.30
9	5.45	1.4519	30.80	0.77	64.95
10	4.8	1.4499	30.90	0.75	65.45
11	5.5	1.4531	30.95	0.79	65.45

Summary on the results of physical test of the different oil composition as presented above shows that all treatments complied to the given standards as to refractive index, density, pH, and brix index. However, although the results on the lower pH can increase the solubility of oils, only T11 (pH = 5.5) is within the skin's pH range of 5.5.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

This study showed that the extraction and the process of producing oil from oregano, lemongrass, neem and butterfly pea is acceptable within the standards based on the results of some physical test in this study. All treatments complied to the given standards as to refractive index, density, pH, and brix index. Refractive index ranges from 1.450 to 1.453, density ranges 0.70-0.82 g/mL, brix index ranges 64.45 – 65.60%. However, although the results on the lower pH can increase the solubility of oils, only T11 (pH = 5.5) is within the skin's pH range of 5.5. Making T11 (5%oregano, 5% lemongrass, 5%neem, and 5%butterfly pea) the most acceptable formulation so far in the coverage of this study.

It is recommended to conduct further analysis on anti-microbial activity, antioxidant activity, skin irritation test, stability studies. Other physical test that were not covered in this study such as saponification value, viscosity, specific gravity, and acid value should be explored. It is also recommended to conduct studies on the application of the oil composition in other products.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 10

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

2 – IPR Application: Herbal Oil (Oreganum Lemongrass, Neem, and Butterfly Pea) and the Process of Producing thereof

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

N/A

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

Direct Cost

Travel Expenses (Local)	P 500. 00
Supplies and Materials/Raw Materials	P 3, 500.00
Tools	P 10, 000.00
Facility and Equipment Rental Expenses	P - -
Documentation Printing, Photocopying Expenses	P 2,000.00
Total	P 16,000.00

ROI (%) = Net Income /Total Project Cost
= 2,100/16,000 = 13%

Payback Period (Years) = Total Project Cost /Average Annual Profit
= 16,000/5,160
= 3.1

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

Sincere gratitude to my Organic Chemistry (AY 2023-2024) students who served as Laboratory assistants during the conduct of the study.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Alzohairy M. A. (2016). Therapeutics Role of Azadirachta indica (Neem) and Their Active Constituents in Diseases Prevention and Treatment. Evidence-based complementary and alternative medicine: eCAM, 2016, 7382506. <https://doi.org/10.1155/2016/7382506>.

Bencheqroun, H. K., Ghanmi, M., Satrani, B., Aafi, A., & Chaouch, A. (2012). Activité antimicrobienne des huiles essentielles d'Artemisia mesatlantica, plante endémique du Maroc Antimicrobial activity of the essential oil of an endemic plant in Morocco Artemisia mesatlantica. Bulletin de la société Royale des sciences de Liège.

Bolivar, L.T. (2017). Process of Producing an Herbal Oil Composed of Tawa-Tawa (Euphorbia Hirta) with Malunggay (Moringa Oleifera) and Garlic (Allium Sativum). (Philippine Patent PH2/2017/000091). IPOPHL
https://patentscope.wipo.int/search/en/detail.jsf?docId=PH278377956&_cid=P20-LZAMN3-49729-1

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 10 of 10

Bourkhiss, M., Hnach, M., Lakhli, T., Bouachrine, M., Farah, A., Satrani, B., ... & Costa, J. (2012). The fractionation and evaluation of essential oils from the sawdust of Moroccan *Tetraclinis articulata* (Vahl) Masters. *Phytothérapie*, 10, 222-228.

Doloriel, D.M, & Doloriel, N.S. (2018). Formulation of An Anti-Fungal Herbal Oil from Katakataka (Bryophyllum Pinnatum Lam.) Oken) Leaves and Makabuhay (Tinospora Crispa L. Hook. F. and Thomson) Stema. Philippine Patent PH 2/2018/000704. IPOPHL https://patentscope.wipo.int/search/en/detail.jsf?docId=PH324270553&_cid=P20-LZAMN3-49729-1

FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS WORLD HEALTH ORGANIZATION Rome (2001). THE CODEX ALIMENTARIUS. Vol.8 2nd Edidtion. Retrieved from <https://www.fao.org/4/y2774e/y2774e00.htm>

Hikmah, N. (2017). Yield and quality of essential oil from ginger (*Zingiber officinale* Rose). *Hutan Tropis Borneo* 8:20,8-16

International Gem Society (IGS). (2024). Refractive Index List of Common Household Liquids. Retrieved from <https://www.gemsociety.org/article/refractive-index-list-of-common-household-liquids/>.

Jeyaraj, E. J., Lim, Y. Y., & Choo, W. S. (2021). Extraction methods of butterfly pea (*Clitoria ternatea*) flower and biological activities of its phytochemicals. *Journal of food science and technology*, 58(6), 2054–2067. <https://doi.org/10.1007/s13197-020-04745-3>

Lourenço, S. C., Moldão-Martins,M. & Alves, V.. (2019). "Antioxidants of Natural Plant Origins: From Sources to Food Industry Applications" *Molecules* 24, no. 22: 4132. <https://doi.org/10.3390/molecules24224132>

Mieso B, Befu A (2020) Physical Characteristics of the Essential Oil Extracted from Released and Improved Spearmint Varieties, Peppermint and Japanese Mint. *Med Aromat Plants (Los Angeles)* 9: 355. doi: 10.35248/2167-0412.20.9.355

Santos, MA. (2018). Herbal Oil. (Philippine Patent PH 2/2018/000019). IPOPHL https://patentscope.wipo.int/search/en/detail.jsf?docId=PH278351932&_cid=P20-LZAMN3-49729-1.

Shah, G., Shri, R., Panchal, V., Sharma, N., Singh, B., & Mann, A. S. (2011). Scientific basis for the therapeutic use of *Cymbopogon citratus*, stapf (Lemon grass). *Journal of advanced pharmaceutical technology & research*, 2(1), 3–8. <https://doi.org/10.4103/2231-4040.79796>

Sharififard, M. et al (2018) Chemical Composition and Repellency of *Origanum vulgare* Essential Oil against *Cimex lectularius* under Laboratory Conditions. *J Arthropod Borne Dis.* 2018 Dec; 12(4): 387–397.

Sharmeen, J. B., Mahomoodally, F. M., Zengin, G., & Maggi, F. (2021). Essential Oils as Natural Sources of Fragrance Compounds for Cosmetics and Cosmeceuticals. *Molecules (Basel, Switzerland)*, 26(3), 666. <https://doi.org/10.3390/molecules26030666>